Analysis Based on Onboard Lamp and Lunar Vicarious Calibrations for Sensitivity Degradation of a Hyperspectral Sensor

Satoru Yamamoto, Hiroki Mizuochi, Moe Matsuoka, Koki Iwao, and Satoshi Tsuchida

***Abstract*— We assessed the degradation of the radiance lamp in the onboard calibration system of the Spectral Profiler, a hyperspectral sensor onboard the Japanese Selenological and Engineering Explorer (SELENE), which was launched in September 2007 and observed the Moon until June 2009. We found that the output of the radiance lamp soon after launch increased by 3.2%–4.6% as compared with the prelaunch output, after which it decreased over time. The total decrease during the mission lifetime of about 1.5 years after the increase of 3.2%–4.6% was ∼1.1%–1.5%, which is slightly larger than the <1% degradation of the sensitivity of the sensor, as derived from vicarious calibration using the Apollo 16 landing site observations. Thus, we confirmed that the radiance lamp became degraded, but the degree of degradation was within 1% during ∼1.5 years. We also found a wavelength dependence in the degradation, in which degradation was larger for shorter wavelengths, especially for wavelengths less than ∼0.8 μm. The trend of the wavelength dependence was shown to be consistent with a model of blackbody radiance with a decrease in temperature of the lamp of about −5 K, suggesting that the degradation of the lamp was mostly due to temperature decrease of about −3 K/yr. Our results may provide an important constraint on the quantitative assessment of the degradation of the sensitivity of remote-sensing sensors using an onboard calibration system, such as the Advanced Spaceborne Thermal Emission and Reflection Radiometer.**

***Index Terms*— Hyperspectral sensor, onboard radiance lamps, SELENE, sensor degradation.**

I. Introduction

ONBOARD calibration systems for remote-sensing instruments are used to evaluate temporal changes in the sensitivity of the sensor (hereafter, the sensor sensitivity) in space, both for Earth observations and planetary missions [1]–[15]. The temporal change in the sensor sensitivity from the monitoring of onboard lamps has been reported for several optical sensors (e.g., Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER) [9], thematic mapper (TM) and multispectral scanner (MSS) onboard Landsat series [2], [3], and Advanced Visible Near Infrared Radiometer (AVNIR) onboard the Advanced Earth Observing Satellite [4]). To use the onboard lamps to assess the degradation of a sensor, we need to separate the change in the output of the lamp (hereafter, the lamp output) due to lamp degradation from the change in the sensor sensitivity [3], [5]. However, it has been unclear how the lamp output changes in space. For example, although the lamp output for ASTER was evaluated by monitoring photodiodes set around the lamps [12], we cannot rule out the possibility that degradation of the photodiodes also occurred, obscuring the degradation of the lamp. The previous calibrations of ASTER and Landsat MSS using onboard lamps assumed a stable lamp output in comparison to the changes in the sensor sensitivity [2], [9].

In this study, we attempt to evaluate the degradation of the onboard lamps for the Spectral Profiler (SP), a hyperspectral remote-sensing sensor onboard the SELenological and ENgineering Explorer (SELENE or Kaguya) spacecraft, which was launched in September 2007 and observed the Moon until June 2009 [16]. While the SP had an onboard calibration system using two halogen lamps, the stability of the SP sensor sensitivity has been evaluated from vicarious calibration using observations of the Apollo 16 landing site (hereafter, A16 observations) at four different times [19], [20]. This location has been widely used as an optical standard in previous Earth-based telescopic observations and various sensors [21]. Note that the vicarious calibration for the Moon have the advantage of evaluating the temporal change of the sensor sensitivity because the lunar surface remains geologically stable over the time scale of human space exploration. According to [19], the difference in reflectance among these four observations was less than the uncertainty of the SP of about 1% for most of the bands, suggesting that degradation of the SP was not significant over the mission period. In this case, the comparison between the onboard lamp and the vicarious calibration allowed us to assess the degradation of the lamp. If the temporal change of the lamp output on the SP was larger than about 1%, we could confirm the upper limit of the degradation of the lamp during the life of the mission. However, if the temporal change of the lamp output is within about 1%, we can consider that the lamp did not degrade significantly during the mission period.

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Satoru Yamamoto, Hiroki Mizuochi, Koki Iwao, and Satoshi Tsuchida are with the Geological Survey of Japan, National Institute of Advanced Industrial Science and Technology, Tsukuba 305-8567, Japan (e-mail: yamamoto.satoru@aist.go.jp).

Moe Matsuoka is with LESIA, Observatoire de Paris, 92195 Meudon, France.

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TABLE I
DETECTOR CHARACTERISTICS OF THE SP

Sub system	Detector	Coverage of λ range [nm]	Sampling of λ interval [nm]	Range of pixel number n
VIS	Si p-i-n photodiode	512.6–1010.7	6	1 to 84
NIR 1	InGaAs	883.5–1676.0	8	85 to 184
NIR 2	InGaAs	1702.1–2587.9	8	185 to 296

Because no radiometric calibration relating to the onboard lamp data for the SP has been reported so far, the temporal change of the lamp output of the SP has been unknown. In this study, from the analysis of the onboard lamp data for the SP, we conducted an assessment of the temporal change in the lamp output and examined whether the lamp became degraded during the mission period. Based on the results, we will also discuss the relevance of our data to other projects and instruments, as well as further interpretation of the SP.

II. OVERVIEW OF SP AND ONBOARD CALIBRATION SYSTEM

In this section, we mainly provide a brief overview of the onboard calibration system related to the SP; a more detailed description of the SP is given in [11], [19], and [20]. The SP is not a hyperspectral sensor imager, but rather a Cassegrain line spectral profiler measuring the spectrum of each footprint with sizes of ∼500 m ×500 m. The SP consisted of one reflective fore-optics unit, two gratings, and three linear-array detectors (VIS, NIR 1, and NIR 2). The light focused by the fore-optics passed through a narrow slit, followed by separation by two visible light and infrared spectrometers (SP–VIS and SP–NIR spectrometers). The light sent to the SP–VIS spectrometer was detected by the VIS detector, which covered visible wavelengths of $\lambda = 512.6$–1010.7 nm. The light sent to the SP–NIR spectrometer was split by a dichroic mirror into the NIR 1 and NIR 2 detectors, which covered near-infrared wavelengths with $\lambda = 883.5$–1676.0 nm and $\lambda = 1702.1$–2587.9 nm, respectively. The detector characteristics are summarized in Table I. Because the error of NIR 2 in the radiometric calibration is estimated to have been 1.0%–2.2% [20], which is larger than that the stability of the sensor sensitivity of <1% based on vicarious A16 observations, we did not use the data of NIR 2 in this study. However, we could use the VIS and NIR 1 data, whose errors in the radiometric calibration are estimated to be ∼0.2% and ∼0.4%–0.7%, respectively [19].

The SP onboard calibration system consisted of two lighting modules attached to the SP fore-optics and a light source module in the spacecraft with two halogen lamps: a radiance lamp and a wavelength lamp. The radiance and wavelength lamps were used for evaluating the radiance degradation and the wavelength shift, respectively. In the onboard calibrations using the lamps, we use the pre-launch data as a reference, but we focus on how much the radiance degrades with time in space. Thus, the lamps were not subjected to any scrutiny to see how well the lamps at pre-launch tests meet the original product specifications. The two modules were connected by optical fibers, whose one side was bare and other side was equipped with a collimator, and the light from the lamps passed through optical fibers to the fore-optics. This means that the path of the light from the lamp was not the same as the actual path of the observed data through the full optical system of the instrument. Except for the optical fibers used in the lamps, there was no additional optical component in the lamp lighting system. The power of the halogen lamps was 9 W, and the sources were stabilized in voltage. When monitoring the voltage from telemetry data, we did not find any change in the voltage during the mission. There was no heat run mode for the lamp calibration. Data acquisition started a few seconds to ∼10 s after the lamp was turned on. The filament temperature in the product standard (ROYAL; HLG-SE6-545) was 2700 K. The total usage time of the lamps during the mission is ∼105 h, which is much shorter than the expected lamp life of 5000 h.

While the main optical experiment of the SP, including the determination of the radiometric calibration coefficients (RCCs), was conducted at the Japan Aerospace Exploration Agency (JAXA), Tsukuba Space Center, the pre-launch data for the radiance lamp and the wavelength lamp were measured at the Tanegashima Space Center in Japan [18], [19]. The pre-launch data were obtained under atmospheric pressure. In addition, the lamps were not seasoned or baked in before installing them in the instrument and during the pre-launch tests. In this case, there is the possibility that water vapor affected the pre-launch data for the radiance and the wavelength lamps. Indeed, it has been reported that several features probably due to water vapor appear in the observational data of the SP. The detailed analysis with the radiative transfer code (MODTRAN 4) showed that the radiances in $\lambda = 1300$–1400 nm and 1800–1950 nm could be affected by water, although no clear water features were observed in other wavelength ranges [19], [20]. On the other hand, as mentioned below, we will not use the data in $\lambda = 1300$–1400 nm in the following analysis.

Fig. 1 shows S-values of the pre-launch data of the radiance and wavelength lamps as a function of pixel number n. For the S-value, S is defined as

$$S = D_{\text{ori}} - D_{\text{drk}} \tag{1}$$

where D_{ori} and D_{drk} are the original digital number (DN) and background/dark DN, respectively. While the S-values for the radiance and the wavelength lamp have a similar spectral shape, the later exhibits several absorption features, especially in the VIS range (e.g., $n = 12$–13, 39–40, 50, and 62–63), which allows us to evaluate the wavelength shift.

Finally, we briefly explain the A16 observations as follows. Four observations for the Apollo 16 landing site were conducted at 6-month intervals: November 19, 2007,

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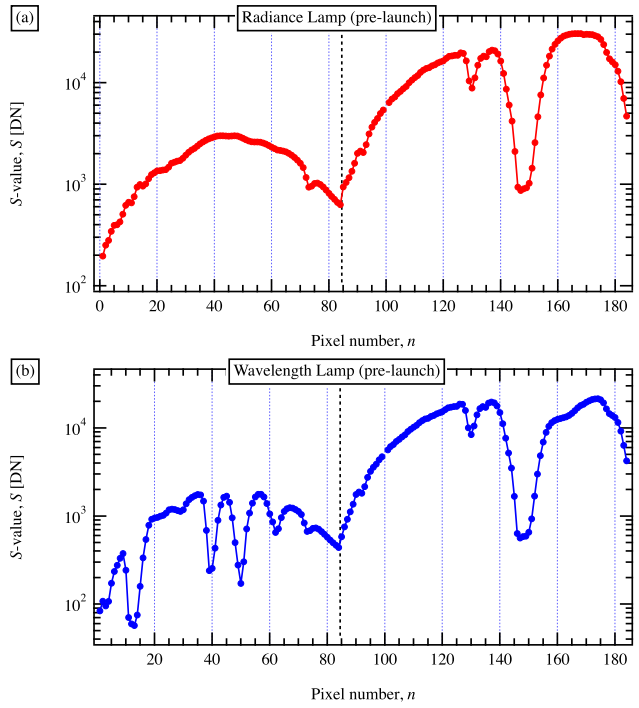


Fig. 1. (a) *S*-values of the prelaunch data as a function of the pixel number n for (a) radiance lamp and (b) wavelength lamp. The vertical dotted line indicates the boundary between VIS ($n \leq 84$) and NIR 1 ($n \geq 85$).

March 21, 2008, September 15, 2008, and March 12, 2009. In each observation, SELENE tilted its attitude (roll-cant maneuver) to attain the condition with a phase angle of ~30°. Although this could mitigate the effects of the different solar illumination, there remain some differences in the data. Therefore, photometric corrections were also taken into account in the analysis of the A16 observations. This also includes the correction of the difference in the sun–moon distance. As a result, the difference in reflectance after the photometric correction has been estimated to be 0.4%–1.1% [19].

III. Analytical Procedure

According to [19] and [20], stray light was found in the SP observation data immediately after SELENE entered the shadow of the Moon. In addition, a wavelength shift with short time variation was observed in the SP data. Therefore, it is also expected that the lamp data analyzed in this article may have been affected both by the stray light and the wavelength shift, which would greatly affect the assessment of the ratio of the *S*-value between the in-flight data and the pre-launch data. Thus, we first describe the analytical procedures developed to avoid or correct the data affected by the stray light and the wavelength shift for the lamp data.

A. Stray Light

Fig. 2 shows a schematic of eight positions of SELENE, (a)–(h), during one revolution around the Moon. In the near-polar orbit of SELENE, while normal observation was conducted on the dayside of the Moon [e.g., position (a)], the onboard calibrations with the radiance and wavelength

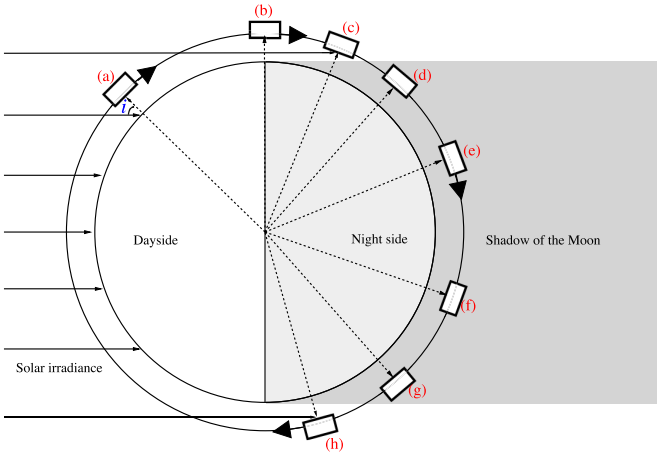


Fig. 2. Schematic of the positions of SELENE and the Moon, where the altitude and size of SELENE are exaggerated. Parameter i is the solar incident angle ($i = 0°$ corresponding to the direction normal to the surface). This is the case of $\beta = 0°$, where the β angle is defined as i at the equator on the dayside.

lamps were conducted on the night side of the Moon [e.g., positions (e) and (f)]. The dark DN for the onboard calibration was also measured before [from position (c) to (d)] and after each onboard calibration on the night side [from positions (g) to (h)]. It should be noted that the stray light was observed at positions (c) and (h) because there is a time range during which the SELENE body was still outside the shadow of the Moon but the bottom side of the SELENE body was tilted toward the Sun. Because the SP did not have a shutter to cut incident light, the sunlight hitting the bottom side of SELENE reached the spectrometers through the fore-optics, manifesting as stray light in the SP data (more details are given in [19] and [20]).

Because the measurements of the dark DN were conducted when the solar incident angle i was $\geq 90°$ without taking accounting for stray light, some data in the onboard calibrations were affected by stray light. To assess which data were affected by the stray light, we estimated the critical solar incidence angle i_c at which SELENE first entered the shadow of the Moon as follows. Assuming that the Moon has a spherical shape with a radius of R_M and that the SELENE orbit described a perfect circle with a height H from the lunar surface, we can estimate i_c from the following relation:

$$\sin(\pi - i_c) = \frac{R_M}{R_M + H}. \tag{2}$$

From (2), we estimate $i_c = 109.0°$ for the nominal planned height of $H = 100$ km and $i_c = 111.5°$ for the maximum height of $H = 130$ km, suggesting that the SP data in i ranging from 90° to ~112° would be affected by the stray light. Fig. 3 shows the D_{drk} values that were affected by the stray light at $\Phi = 3345$ with $i = 91.6°$ to 92.1°, and the values that were not affected by the stray light at $\Phi = 2007$ with $i = 136.8°$ to 141.4°, where Φ is the total number of revolutions of SELENE around the Moon. We can see that there is a peak centered around $\lambda \sim 750$ nm for $\Phi = 3345$, while the data for $\Phi = 2007$ does not have this peak. (The peak around $\lambda \sim 750$ nm

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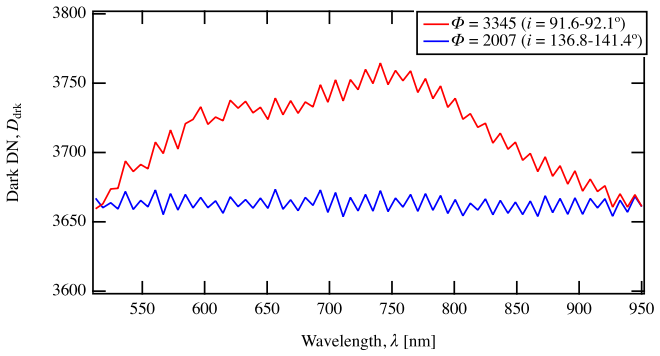


Fig. 3. Comparison of the dark DN (D_{drk}) for VIS between $\Phi = 3345$ ($d = 299$ days) with $i = 91.6°$ to $92.1°$ and $\Phi = 2007$ ($d = 189$ days) with $i = 136.8°$ to $141.4°$.

is due to the response characteristics of the VIS sensor to the input light [19].) Because the pre-launch data did not show such a peak in the D_{drk} curve, it is obvious that the data with $\Phi = 3345$ are affected by stray light.

Note that the schematic in Fig. 2 is the case of $\beta = 0°$, where β is defined as i at the equator at the dayside. For the case of $\beta = 0°$, the orbital plane of SELENE is vertical to the Sun, and SELENE could reach the far side of the moon's night side with $i \sim 180°$. In contrast, for larger β, the orbital plane was closer to horizontal to the Sun, and SELENE orbited around the terminator between the day and night sides. In this case, both D_{drk} and the wavelength/radiance lamp observations were affected by the stray light with $i < 112°$. In Fig. 4(a) and (b), the values of i at the start and end of the wavelength/radiance lamp observations and the dark DN observations are plotted against the number of days d after launch. We can see that there are three periods with $i < \sim 112°$ (below the green line in Fig. 4): 1) $d = 89$ to 141 (88 to 140); 2) 264 to 288 (266 to 326); and 3) 453 to 509 (452 to 509) for the wavelength lamp observations (with radiance lamp observations in parentheses). Furthermore, there is a possibility that the temperature of the spectrometer was not stable immediately after SELENE entered the shadow of the Moon, affecting the wavelength/radiance lamp observations. Therefore, we used only the data with $i > 120°$ as a conservative estimate in the following analysis.

B. Correction for Wavelength Shift

We next examined the wavelength shift based on the wavelength lamp (hereafter, WL data), where the wavelength shift is defined as the deviation of the center wavelength for each pixel from the pre-launch data. Because it has been reported that the behavior of the wavelength shift for the SP observational data is different between VIS and NIR 1 [19], we examined VIS and NIR 1 for the lamp data separately.

1) VIS: According to [19], the amount of the wavelength shift $\delta\lambda$ in VIS showed a rapid variation during one orbit of SELENE around the Moon, where $\delta\lambda$ was evaluated using the peak positions of the wavy pattern due to the inherent spectral feature of the low-pass filter used to determine the RCC in the pre-launch experiments. To correct the wavelength shift, [19] proposed an empirical relation as a function of the

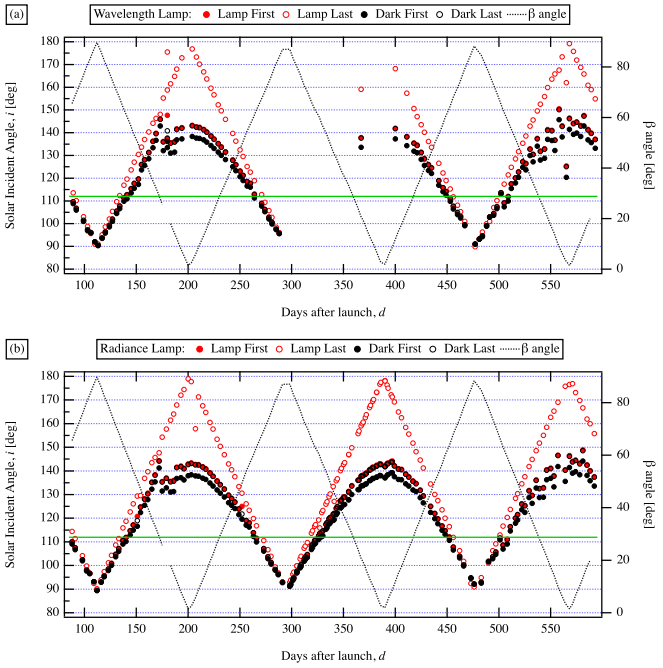


Fig. 4. Solar incident angle i at the starting and ending times of the onboard calibration and the dark DN is plotted against days d after launch for (a) wavelength lamp observation and (b) radiance lamp observation. The starting time and ending time to obtain dark DN data are indicated as solid black circles ("Dark First") and open black circles ("Dark Last"), respectively. The starting and ending times to conduct wavelength/radiance lamp observations are indicated as solid red circles ("Lamp First") and open red circles ("Lamp Last"), respectively. The dotted line denotes the β angle. The horizontal green line denotes $i = 112°$.

temperature of the spectrometer, in which the error in the estimate by the empirical relation was estimated to be within $\sim$0.3 nm. Therefore, applying the empirical relation to the WL data is expected to produce no significant wavelength shift in the VIS data. Nevertheless, we found that there is a considerable amount ($|\delta\lambda| \sim 2$ nm) of wavelength shift in the S-values even after correction using the empirical relation. Fig. 5(a) shows a comparison in S-value between the WL data at $\Phi = 2273$ and the pre-launch data, where the WL data are corrected using the empirical relation. It is clear that the S-value at $\Phi = 2273$ is shifted to the left in the pre-launch data. Using the same method to quantify the wavelength shift based on the wavy pattern as that used in [19], we estimated $\delta\lambda = -2.3$ nm.

We also estimate $\delta\lambda$ for the WL data at various d in Fig. 5(b), where $\delta\lambda$ for VIS is mostly constant at about -2.3 nm, suggesting a nearly constant offset in $\delta\lambda$ in VIS. The offset of $\delta\lambda$ of about -2.3 nm is probably due to the difference in the temperature condition between the two pre-launch experiments, because the pre-launch experiments to obtain the RCC were conducted at JAXA Tsukuba Space Center, but the pre-launch experiments to obtain the WL data were performed at Tanegashima Space Center [18], [19].

Thus, in this study, we applied the additional wavelength-shift correction of $\delta\lambda = +2.3$ nm for all the radiance lamp data.

2) NIR 1: Contrary to the VIS results, the radiometric calibration for the observational data in [19] did not account

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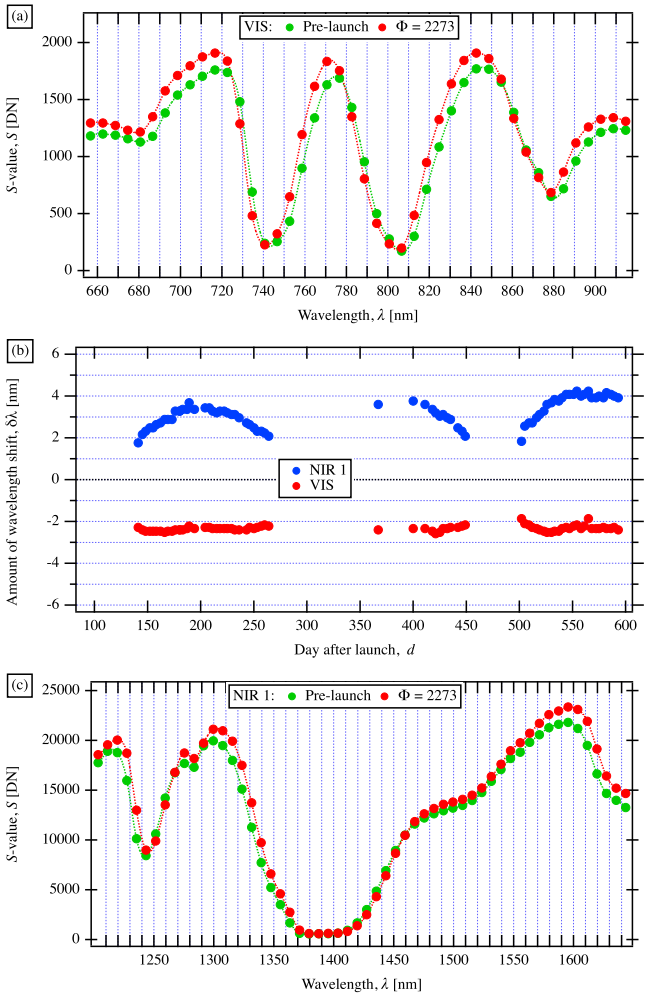


Fig. 5. (a) Comparison of the S-value between the WL data at $\Phi = 2273$ ($d = 211$ days) (red circles) and the prelaunch data (green circles) at VIS wavelengths. The dotted curves are cubic spline interpolations to determine how well the S-value shows the wavelength shift. (b) Wavelength shift $\delta\lambda$ for VIS (red circles) and NIR 1 (blue circles) at various d, where we omit the data with $i < 120°$. (c) Same as (a), but at NIR 1 wavelengths.

for the wavelength shift in NIR 1 because the wavelength shift in NIR 1 was reported to be within $\delta\lambda\sim2$ nm, which was inferred from a small depression feature in the RCC of NIR 1. However, from the WL data, we found wavelength shifts at $\delta\lambda$ = 2–4 nm. Fig. 5(c) shows a comparison of the S-values between the WL data at $\Phi = 2273$ and the pre-launch data. It is clear that the S-value at $\Phi = 2273$ is shifted to the right in the pre-launch data. The direction of the deviation of NIR 1 is opposite that of VIS even for the same Φ, because the detector and optical system for NIR 1 is different from those for VIS. Using the same method to quantify the wavelength shift based on the wavy pattern as that used in [19], we estimated $\delta\lambda = 3.2$ nm for Fig. 5(c).

We then estimated $\delta\lambda$ for the WL data in NIR 1 data at various d, as shown in Fig. 5(b), where we can see that $\delta\lambda$ varies with d, probably due to the variation of the temperature of the spectrometer related to the β angle. Note that the radiance lamp data in NIR 1 have a clear wavy pattern, as shown in Fig. 1(a). Thus, in this study, we corrected the wavelength shift in each set of radiance lamp data for NIR 1 based on the same method using the wavy pattern as that in [19].

3) Simulation of Wavelength Shift: The error in the estimate by the empirical relation was reported to be $\delta\lambda\sim0.3$ nm for the correction of the wavelength shift based on the wavy pattern used in [19]. While the error with $\delta\lambda\sim0.3$ nm would not be significant for spectral identification of major lunar minerals [22]–[28], even a tiny wavelength shift with $\delta\lambda\sim0.3$ nm would affect the ratio χ of S_{in} to S_{pr} as

$$\chi = \frac{S_{\text{in}}}{S_{\text{pr}}} \tag{3}$$

where S_{in} and S_{pr} are S-values of the in-flight data and the pre-launch data, respectively. The ratio of S_{in} to S_{pr} corresponds to the output ratio of the lamp between the in-flight and preflight data (hereafter, we call this the S-ratio), and the S-ratio is used to examine how the lamp output for the in-flight data changes as compared to the pre-launch data. To see how the wavelength shift with $\delta\lambda\sim0.3$ nm affects the S-ratio, we simulated the wavelength shift using the pre-launch data S_{pr}. In this simulation, we first interpolated S_{pr} using the cubic spline interpolation (shown by dotted curves in Fig. 5). Then, we calculated the S-ratio for the shifted S-value with a given $\delta\lambda$. In Fig. 6(a), the S-ratio is plotted against n for the cases of $\delta\lambda = 0.1$, 0.3, and 0.5 nm. We can see that even for $\delta\lambda = 0.1$ nm, wavy patterns, which are due to the difference in the peak position between the shifted S-value and S_{pr}, are observed at several pixels. It is also clear that the amplitude of the wavy patterns increases with increasing $\delta\lambda$.

Fig. 6(b) shows the absolute error E_{wv} due to the wavelength shift, where the absolute error is defined as $E_{\text{wv}} = |\chi - 1|$. We can see that while E_{wv} increases with increasing $\delta\lambda$ at most of the pixels, E_{wv} at several pixels remains less than 0.2% (e.g., n = 42–47, 54–55, or 165–171). In other words, there are two types of pixels in terms of the sensitivity to the wavelength shift: 1) pixels sensitive to even small $\delta\lambda$ (hereafter, sensitive pixels) and 2) pixels insensitive to $\delta\lambda$ (hereafter, insensitive pixels). For comparison, we also plot the S-value of the prelaunch in Fig. 6(b), where we can see that the insensitive pixels correspond to the case where the gradient of the S-value of the pre-launch is nearly flat, while the sensitive pixels correspond to the case at which the gradient of the S-value of the pre-launch is steep. This suggests that the sensitive pixels are less suitable for the assessment of the temporal change of the S-ratio, because even a small $\delta\lambda$ of about 0.3 nm can result in a large error in the S-ratio. In addition, the lower S-value would be easily affected by noise components, resulting in a larger error in the S-ratio. The data in λ = 1300–1400 nm (n = 138–149), which could be affected by water vapor, also correspond to the sensitive pixels. Thus, we should use the insensitive pixels with the larger S-value for assessment of the temporal change of the S-ratio. Therefore, in the following analysis, we used the pixels whose $S_{\text{pr}} > 2000$ and $E_{\text{wv}} < 0.2\%$, especially for assessment of the temporal change in the S-ratio.

As seen in Fig. 1, the wavelength lamp has several absorption features, especially at VIS wavelengths, resulting in a

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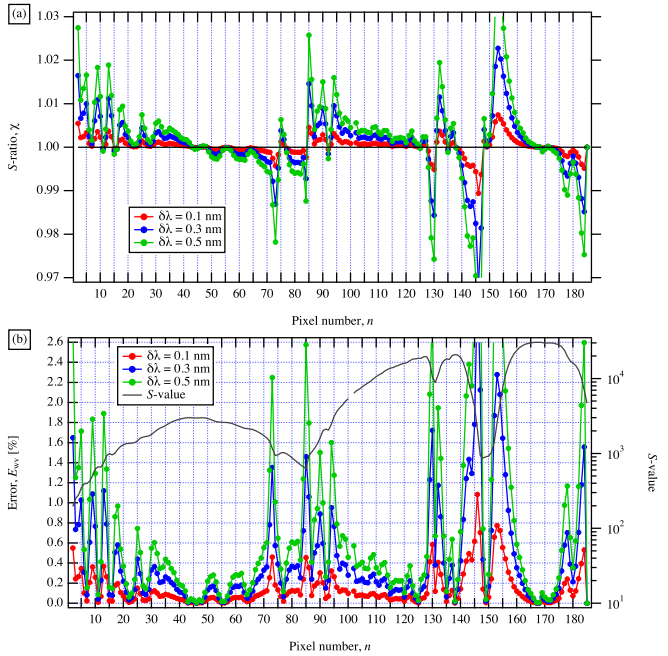


Fig. 6. (a) S-ratio χ of the shifted S-value and the pre-launch data S_{pr} is plotted against n for $\delta\lambda = 0.1$ nm, 0.3 nm, and 0.5 nm. (b) Error $E_{\mathrm{wv}} = |\chi - 1|$ for the results in (a) is plotted against n. For comparison, cubic spline interpolation of S_{pr} is plotted on the right axis.

larger error in the S-ratio as the insensitive pixels. Thus, we do not use the WL data for assessments of the S-ratio in the following analysis.

C. Nonlinearity Correction and Out-of-Range Response Pixels

Following the method of [19], we applied the nonlinearity correction to S-value after the wavelength corrections as

$$S_{\mathrm{c}} = S + kS^2 \tag{4}$$

where S is the S-value before the nonlinearity correction, S_{c} is the S-value after the nonlinearity correction, $k = 9.751 \times 10^{-7}$ for VIS, and $k = 6.176 \times 10^{-7}$ for NIR 1 from [19]. In the following, the S-ratio is calculated using S_{c} for the in-flight data and the pre-launch data.

In addition, following [19], [20], we omit the following pixels as an out-of-range response: 1) pixels with $n = 100$, which are known as abnormally responding pixels from the pre-launch experiment, 2) pixels with $n = 75$-93, which correspond to overlapping wavelengths between VIS and NIR 1, and 3) pixels with $n = 181$–184, which are NIR 1 longer-end pixels.

IV. Results

Using the radiance lamp data, we first examined how the S-value, S_{c}, for the in-flight data is different from that for the prelaunch data, and how the S-ratio changes with time. We also see the correlation with the phase of the Moon to check the effect of Earthshine. Next, we investigated the wavelength dependence of temporal changes in the S-ratio, and compared it with the change in the sensor sensitivity of the SP derived from the A16 observations.

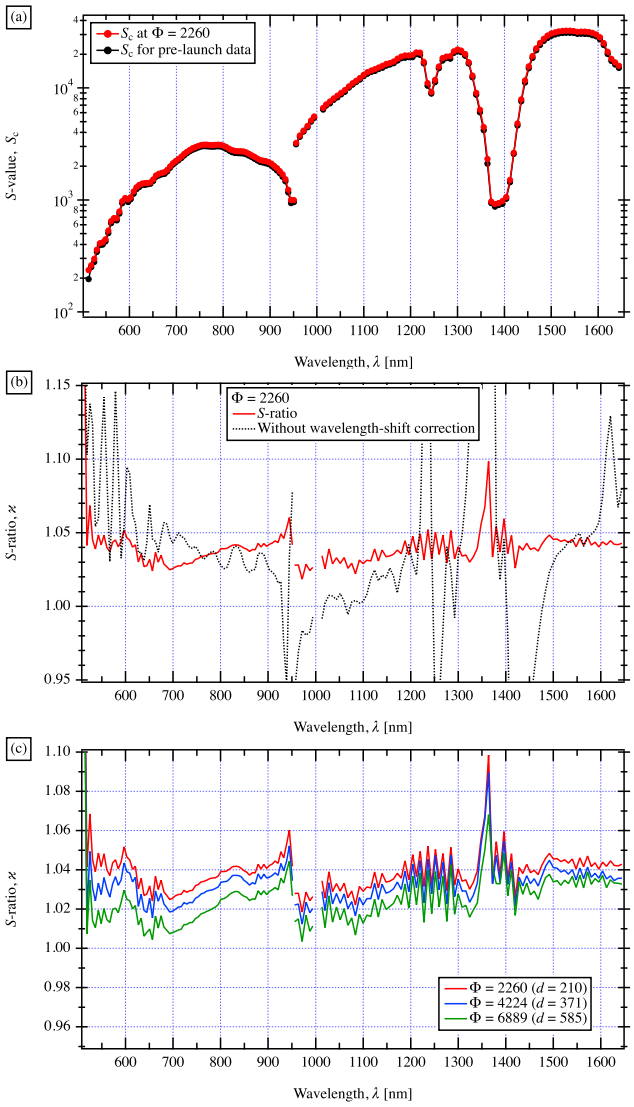


Fig. 7. (a) Comparison of the S-value for the radiance lamp between the in-flight data at $\Phi = 2260$ and the prelaunch data. (b) S-ratio at $\Phi = 2260$ (red curve) is plotted against λ. For comparison, the S-ratio without the wavelength-shift correction (black dotted curve) is also plotted. (c) Comparison of the S-ratio for $\Phi = 2260$ ($d = 210$ days), 4224 ($d = 371$ days), and 6889 ($d = 585$ days).

A. Comparison of S-Value Between In-Flight and Pre-Launch Data

Fig. 7(a) shows an example of the S-value, where the S-values for $\Phi = 2260$ and for the pre-launch data are shown. While the pattern of the S-value at $\Phi = 2260$ seems to be the same as that for the pre-launch data, we can see that the absolute value for $\Phi = 2260$ is slightly larger than that for the pre-launch data, indicating the S-ratio $\chi > 1$. Fig. 7(b) shows the S-ratio for the result of Fig. 7(a), where we can see that the S-ratio for most of the pixels ranges from ~1.03 to 1.05 with a small variation. This indicates that the S-ratio for the radiance lamp increased 3%–5% after launch. The calibration of ASTER by [9] reported a 2%–3% increase in the output of the halogen lamps, which could be due to gravity shifts in the halogen cycle. Thus, the 3%–5% increase of the

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TABLE II
AVERAGE AND DIFFERENCE VALUES OF THE S-RATIO FOR VIS RESULTS

Pixel Number n	Wavelength λ [nm]	S-ratio First period	S-ratio Last period	Difference in S-ratio between first and last periods
42	758.7	1.0324 ± 0.0010	1.0172 ± 0.0004	0.0152
43	764.8	1.0325 ± 0.0010	1.0176 ± 0.0005	0.0149
44	770.7	1.0331 ± 0.0009	1.0184 ± 0.0004	0.0147
45	776.7	1.0336 ± 0.0009	1.0192 ± 0.0003	0.0144
46	782.7	1.0342 ± 0.0010	1.0199 ± 0.0004	0.0143
47	788.8	1.0347 ± 0.0010	1.0205 ± 0.0005	0.0142
54	830.8	1.0395 ± 0.0011	1.0285 ± 0.0005	0.0110
55	836.8	1.0395 ± 0.0010	1.0285 ± 0.0005	0.0111

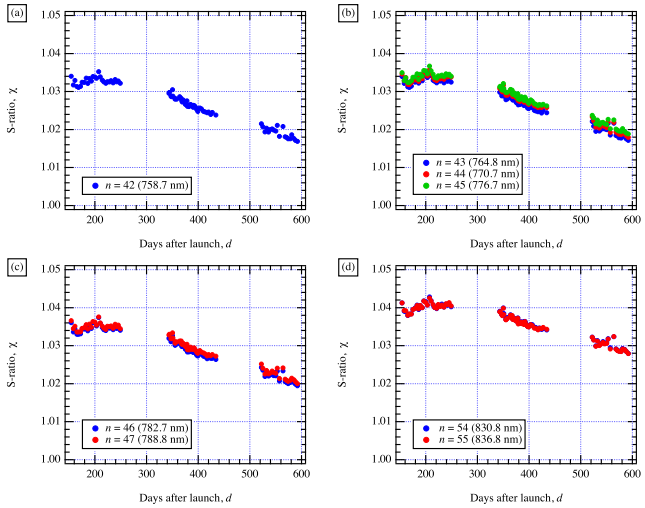


Fig. 8. (a) S-ratio is plotted against d at $n = 42$ ($\lambda = 758.7$ nm). (b)–(d) Same as (a), but for (b) $n = 43$, 44, and 45 ($\lambda = 764.8$, 770.7, and 776.7 nm). (c) $n = 46$ and 47 ($\lambda = 782.7$ and 788.8 nm); and (d) $n = 54$ and 55 ($\lambda = 830.8$ and 836.8 nm).

radiance lamp is also partly due to the gravity shifts in the halogen cycle, as well as the change of the sensor sensitivity of the SP during the launch.

For comparison, we also plot the S-ratio without the wavelength-shift correction in Fig. 7(b) (dotted curve). It is clear that the S-ratio without the wavelength-shift correction shows a large wavy pattern, which is similar to that shown in Fig. 6(a).

We then compare the S-ratio among the three different periods at $\Phi = 2260$ ($d = 210$), 4224 ($d = 371$), and 6889 ($d = 585$) in Fig. 7(c). We can see that the curve shapes for the S-ratio are similar to each other, indicating that the change in the shape of the S-ratio may be due to the change in the sensitivity at each n after launch. Moreover, it is clear that the absolute value of the S-ratio decreases with increasing d. Therefore, the result in Fig. 7(c) indicates a temporal decrease of the S-ratio for the radiance lamp in space.

B. Temporal Change in S-Value for the Radiance Lamp

We next examined how the S-ratio decreases with d. Fig. 8(a) shows an example of the temporal change of the S-ratio for VIS, where the S-ratio at $n = 42$ ($\lambda = 758.7$ nm) is plotted against d. We can see that the S-ratio is nearly constant at 1.030–1.035 for $d < 250$ days, indicating that the S-value in the first period for $d < 250$ days is ~3.0%–3.5% larger than that of the pre-launch. In contrast, it is shown that the S-ratio after $d = 343$ days decreases with increasing d and falls to ~1.017 at d~600 days. The same trend can be observed at other wavelengths, as seen in Fig. 8(b)–(d), where we can see that the S-ratio is nearly constant until $d = 250$ days, followed by a temporal decrease after $d = 343$ days. In Table II, we summarize the average S-ratio for the first period ($d < 250$) and the last period ($d > 585$) for the results shown in Fig. 8. Although there are no data for the S-ratio immediately after launch ($d < 150$), the nearly constant trend in the first period may indicate that there is no large change in the S-ratio for $d < 150$. If so, we may conclude that the S-value for the radiance lamp increased by at least ~3.2%–4.0% after launch, and the total decrease from the first period to the last period was ~1.1%–1.5%.

Fig. 9 shows an example of the temporal change of the S-ratio for NIR 1, where the S-ratio at $n = 121$, 139, 165, and 166 ($\lambda = 1171.8$, 1315.9, 1523.8, and 1531.7 nm) are plotted against d. The S-ratio for $d < 250$ are 1.038–1.042 ($n = 121$), 1.034–1.038 ($n = 139$), 1.044–1.048 ($n = 165$), and 1.042–1.047 ($n = 166$), suggesting that the S-value in the first period is ~3%–4% larger than that before launch, which is consistent with the result for VIS. We can also see that while the decrease in the S-ratio may be small or negligible until d~200 days, the S-ratio decreases with increasing d after $d = 343$ days. The S-ratio at d~600 days are 1.026 ($n = 121$), 1.022 ($n = 139$), 1.034 ($n = 165$), and 1.033 ($n = 166$). In Table III, we summarize the average S-ratio for the first period ($d < 250$) and the last period ($d > 585$) for the results shown in Fig. 9. As with VIS, we can conclude that the S-value for the radiance lamp increased at least by ~3.5%–4.6% after launch, and the total decrease from the first period to the last period was ~1.1%–1.3%.

C. Correlation With Phases of the Moon

Because the lamp observations were conducted on the night side of the Moon, there is a possibility that Earthshine affects the S-ratio as stray light, where Earthshine is defined as an illumination of the Moon by light reflected from the Earth. If this is the case, we would see an intermittent increase in the S-ratio, being correlated with the phase of the Moon, especially around the new moon. To examine the effects of Earthshine, we investigated how the S-ratio varies with the

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TABLE III
AVERAGE AND DIFFERENCE VALUES OF S-RATIO FOR NIR 1

Pixel Number n	Wavelength λ [nm]	S-ratio First period	S-ratio Last period	Difference in S-ratio between first and last periods
121	1171.8	1.0389 ± 0.0011	1.0265 ± 0.0005	0.0123
139	1315.9	1.0352 ± 0.0008	1.0221 ± 0.0003	0.0131
165	1523.8	1.0455 ± 0.0010	1.0343 ± 0.0004	0.0112
166	1531.7	1.0436 ± 0.0012	1.0327 ± 0.0002	0.0109

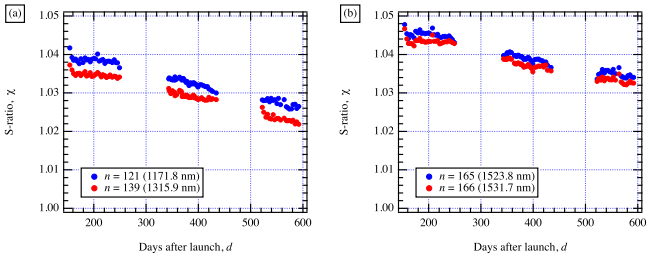


Fig. 9. (a) S-ratio is plotted against d at $n = 121$ and 139 ($\lambda = 1171.8$ and 1315.9 nm). (b) Same as (a), but for $n = 165$ and 166 ($\lambda = 1523.8$ and 1531.7 nm).

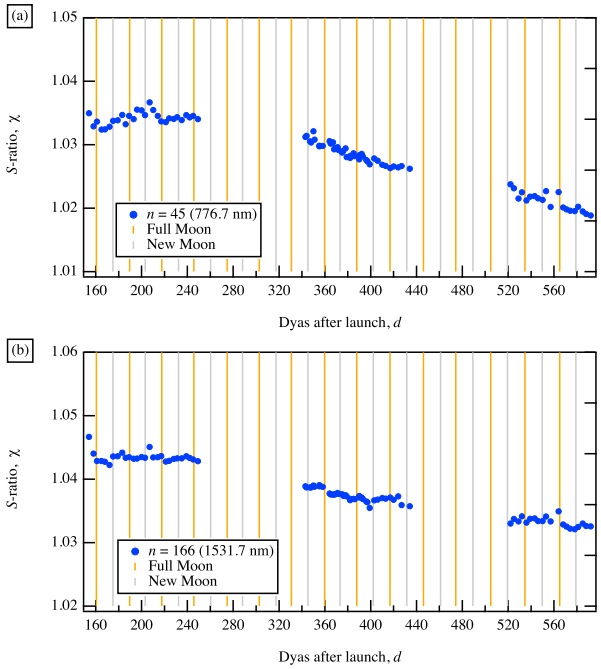


Fig. 10. Comparison between the S-ratio and the timings of the new moon and full moon for (a) $n = 45$ ($\lambda = 776.7$ nm) and (b) $n = 166$ ($\lambda = 1531.7$ nm).

timings of the new moon and full moon in Fig. 10. However, we did not see any clear correlation between the S-ratio and the timings of the new moon and full moon. Thus, we conclude that Earthshine did not affect the S-ratio.

D. Wavelength Dependence of Temporal Change in S-Ratios

In Fig. 11(a), the average S-ratios at four periods of P1, P2, P3, and P4 are plotted against λ, where P1 is defined as $d = 154$–189 days, P2 is $d = 351$–355 days, P3 is $d = 424$–434 days, and P4 is $d = 585$–592 days. (Here, the S-ratio is shown only at λ with $S_c > 2000$ and $E_{wv} < 0.2\%$; see Section III-B3.) We can see that the S-ratio with λ varies during each period, but not in a systematic way. For VIS, while the S-ratio for λ <∼830 nm shows an increase with increasing λ, the S-ratios for λ∼830–900 nm hardly vary with λ. For NIR 1, while the overall trend of the S-ratio for λ∼1000–1200 nm appears to increase with increasing λ, the S-ratios for λ >∼1500 nm hardly vary with λ. The variation with λ may be due to the change in the sensor sensitivity for each pixel after launch.

In contrast, we can see that the absolute S-ratios decrease with increasing d, suggesting a temporal decrease in the S-ratio. In Fig. 11(b), we also show the differences ($\delta\chi_4$, $\delta\chi_3$, and $\delta\chi_2$) in the S-ratio between P1 and various periods, where $\delta\chi_4$ is the difference in the S-ratio between P1 and P4, $\delta\chi_3$ is between P1 and P3, and $\delta\chi_2$ is between P1 and P2. We can see that $\delta\chi_4 > \delta\chi_3 > \delta\chi_2$ for both VIS and NIR 1, indicating that the degree of temporal change in the S-ratio increases with increasing d. In addition, it is shown that the degree of temporal change depends on λ, especially for $\delta\chi_4$ and $\delta\chi_3$ in VIS. Although $\delta\chi_2$ hardly depends on λ for either VIS or NIR 1, $\delta\chi_3$ in VIS increases with decreasing λ, especially for λ <∼830 nm. Moreover, $\delta\chi_4$ in VIS shows a steeper increase with decreasing λ, especially in λ <∼830 nm. This means that the shorter the wavelength, the larger the decrease in the S-ratio, especially for shorter VIS wavelengths. In contrast, we do not see a clear wavelength dependence in $\delta\chi_4$, $\delta\chi_3$, and $\delta\chi_2$ in NIR 1.

For comparison, we also estimate the change (D_{A16}) in the sensor sensitivity of the SP between P1 ($d = 154$-189) and P4 ($d = 585$-592) based on the A16 observations as follows. First, we calculate the ratio R_{A16} of the reflectance of the SP between the second A16 observation ($\Phi = 2008$ and $d = 189$) and the fourth A16 observation ($\Phi = 6375$ and $d = 545$) [19]. Then, assuming a linear proportional model for time, we estimate D_{A16} as

$$D_{A16} = R_{A16} \cdot \frac{\Delta d}{356} \tag{5}$$

where 356 corresponds to the number of days from the second to the fourth A16 observations [19], and Δd is the number of days from P1 to P4. Since there is uncertainty in d for P1 and P4, we estimate the maximum value of D_{A16} for $\Delta d = 438$ days and the minimum one for $\Delta d = 396$ days. In Fig. 11(b), the maximum and minimum values of D_{A16} are plotted. We can see that while $\delta\chi_4$ in $\lambda = 800$–900 nm is comparable to or slightly lower than D_{A16}, the values of $\delta\chi_4$ for $\lambda < 800$ nm and $\lambda > 1000$ nm are larger than D_{A16}. The reason why $\delta\chi_4$ in $\lambda = 800$–900 nm is comparable to or

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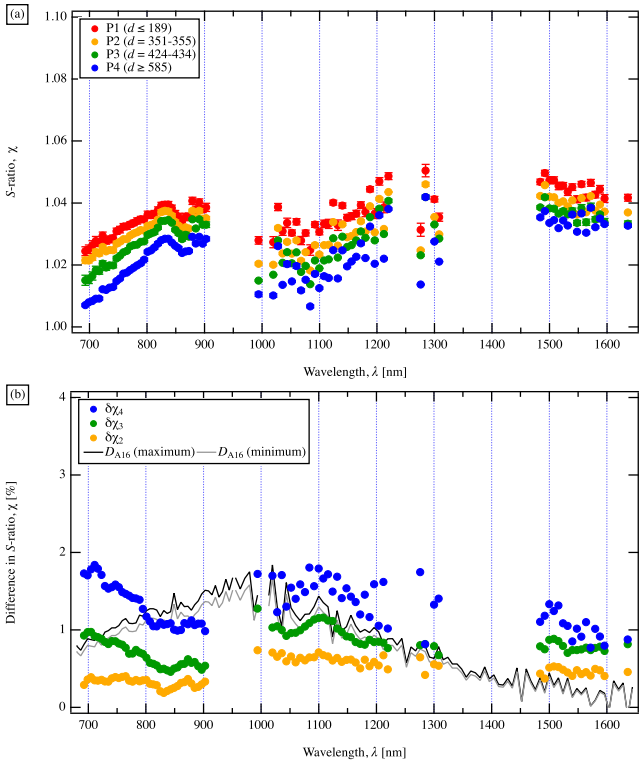


Fig. 11. (a) S-ratio is plotted against λ for periods P1 ($d = 154$–189 days), P2 ($d = 351$–355 days), P3 ($d = 424$–434 days), and P4 ($d = 585$–592 days), where only λ with $S_c > 2000$ and $E_{wv} < 0.2\%$ is shown (see Section III-B3). (b) Differences $\delta\chi_4$ between P1 and P4 (blue circles), $\delta\chi_3$ between P1 and P3 (green circles), and $\delta\chi_2$ between P1 and P2 (orange circles). For comparison, the change D_{A16} in the sensor sensitivity of the SP between P1 and P4 is plotted, where the maximum (black curve) and minimum (gray curve) values are estimated for $\Delta d = 438$ days and $\Delta d = 396$ days, respectively.

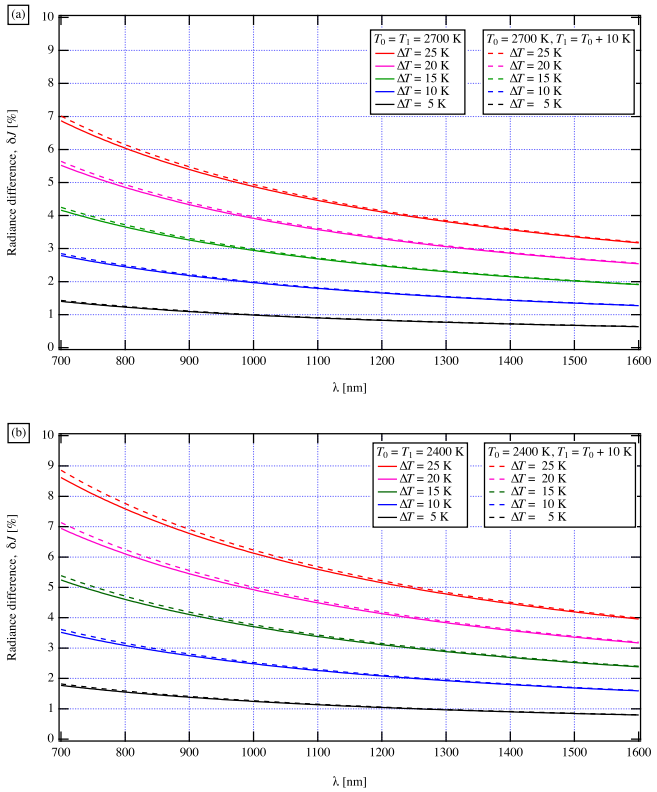


Fig. 12. (a) δJ is plotted against λ for various values of $\Delta T = T_2 - T_1$, where we assume that $T_0 = T_1 = 2700$ K (solid curves) and $T_1 = T_0 + 10$ K (dashed curves). (b) Same as (a), but for $T_0 = 2400$ K.

slightly lower than D_{A16} may be due to a spatial variation in the A16 observation sites conducted by the SP, as discussed below. Thus, the temporal decrease based on the radiance lamp observations is larger than that for the sensor sensitivity of the SP based on the A16 observations, indicating that the radiance lamp became degraded during about 1.5 years. Taking into account the errors of 0.2% (VIS) and 0.4%–0.7% (NIR 1) in the data of the A16 observations [19], we can also estimate that the maximum difference between $\delta\chi_4$ and D_{A16} is $\sim$1%, suggesting that the degree of degradation of the radiance lamp between P1 and P4 could be up to $\sim$1%.

V. Discussion

A. Temperature Decrease of the Lamp

As seen in Fig. 11, we found that the shorter the wavelength, the larger the decrease in the S-ratio, especially for shorter VIS wavelengths. We hypothesize that this can be accounted for by the decrease in temperature T of the lamp, because the peak position of the radiance from the lamp shifts to longer wavelengths with decreasing T, resulting in a larger decrease at shorter wavelengths. To examine this hypothesis, we calculated how the ratio of the radiance between the two different temperatures from the lamp varies with T as follows. Using the radiance $J(T)$ from a blackbody with T calculated from Planck's law, we obtain the difference δJ between $J(T_2)$ and $J(T_1)$ as

$$\delta J = \frac{J(T_1)}{J(T_0)} - \frac{J(T_2)}{J(T_0)} = \frac{\exp\left(\frac{hc}{k\lambda T_1}\right) - \exp\left(\frac{hc}{k\lambda T_2}\right)}{\exp\left(\frac{hc}{k\lambda T_0}\right) - 1} \tag{6}$$

where T_0 is the temperature in the pre-launch data, T_1 and T_2 are the given temperatures at various times, h is the Planck constant, k is Boltzmann's constant, and c is the speed of light. Fig. 12(a) shows δJ for various values of $\Delta T = T_1 - T_2$, where we use $T_0 = T_1 = 2700$ K as the filament temperature of the halogen lamp. We can see that δJ increases with increasing ΔT. In addition, Fig. 12(a) shows a clear wavelength dependence in δJ, where the shorter the wavelength, the larger the change in δJ, which is consistent with the wavelength dependence shown in Fig. 11. In addition, there is a possibility that the temperature of the radiance lamp increased after launch due to the change in gravity. Thus, we also calculated the case of $T_1 = T_0 + 10$ K [dashed curves in Fig. 12(a)], and it shows the same trend as for the case of $T_0 = T_1$. These results support the idea that the temporal decrease in the S-ratio is accounted for by the decrease in T of the radiance lamp. We can also infer that ΔT is about -5 K for $\delta\chi_4$ from Fig. 12.

For a parameter study, we also calculated the case of a lower temperature with $T_0 = 2400$ K [13], [14] in Fig. 12(b). We also see the same trend of the wavelength dependence shown in Fig. 11. It is therefore concluded that the temporal

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decrease in the S-ratio can be mostly accounted for by the decrease in T of the radiance lamp, and ΔT is estimated to be about -5 K during $\sim$1.5 years.

We may also suggest that the decrease in T reflects a decrease in the physical temperature of the filament. What we want to emphasize here is that the wavelength dependence is especially pronounced at $\lambda < \sim$800 nm. For example, as an alternative explanation, the degradation of the lamps might be due to a change of the properties of the filament, that is, simply a shift in the color temperature. In this case, however, it is unknown why a change of the filament properties occurred only for $\lambda < \sim$800 nm. Similarly, we cannot rule out the possibility that only the radiance of the lamps at shorter wavelengths attenuated in the light transmission process owing to contaminants on the optical fiber connecting the halogen lamps to the fore-optics. Nevertheless, there is no particular reason for the wavelength dependence, especially at $\lambda < \sim$800 nm. Therefore, according to the information available at this stage, we consider that this may be due to a decrease in the physical temperature of the filament.

B. Implication for Other Sensors

The wavelength dependence of the temporal change of the S-ratio found in our study may provide valuable interpretations for past calibrations of satellite sensors using onboard lamps. For example, it has been reported that the lamp calibration for the MSS onboard ERTS-1 showed that temporal degradation was observed in bands 1, 2, and 3 with $\lambda < 0.8\,\mu$m, while band 4 with $\lambda = 0.8$–$1.1\,\mu$m was not degraded. The original interpretation [1] proposed that the decrease in the signal was due to spectrally selective contaminants deposited on the scanning mirror and fiber optics. However, it was unclear why the degradation was observed only for $\lambda < 0.8\,\mu$m, and no particular reason for the wavelength dependence was discussed. On the other hand, our results also showed a similar wavelength trend, where the change of the sensor sensitivity in VIS shows a steeper increase with decreasing λ for $\lambda < \sim 0.8\,\mu$m, while that for $\lambda > \sim 0.8\,\mu$m does not. When analyzed together with the A16 observation data, we found that the trend of the wavelength dependence can be accounted for by a decrease in temperature of the radiance lamp. This gives us a possible interpretation that the results of the lamp calibration for the MSS may also be due to a temporal decrease in the temperature of the lamps. If so, this information can be used to improve the design of the next lamp calibration system. For example, we may suggest the importance of simultaneously monitoring changes in temperature and radiance of onboard lamps.

In addition, our analysis and results could provide important information on onboard calibration with lamps for hyperspectral sensors. Although several hyperspectral sensors onboard satellites have been operated in the past, detailed analyses in space using onboard lamps have not been performed well. On the other hand, we performed the high-precision wavelength shift analysis on the hyperspectral data using the lamps and combined it with vicarious calibration from the A16 observations. As a result, we have obtained a variety of new findings of onboard calibration for the hyperspectral data. For example, we found that even a tiny wavelength shift with $\delta\lambda \sim 0.3$ nm affects the ratio of the in-flight data to the preflight data, resulting in a change in the apparent radiance of the lamps. This would also suggest that detailed assessments of wavelength shift are needed for analysis of the sensor degradation using onboard lamps of the Hyperspectral Imager Suite (HISUI), which is a hyperspectral sensor launched in December 2019 and is currently operating on the International Space Station [30]. In addition, our results may also be applied to a long-term degradation analysis using onboard calibration systems of multiband sensors. For example, ASTER has been monitoring sensor sensitivity using onboard lamps for more than 20 years, and the analysis based on onboard lamp calibration and vicarious calibration has reported that there is a band dependence in the trend of the degradation of the sensor [9], [12], [29]. In this case, there is a possibility that the band dependence can be accounted for by the effects of wavelength shift as well as the effects of temperature change of the lamp. Thus, analysis of the ASTER data taking into account the effects of wavelength shift and temperature change of the lamps may provide a better interpretation of the long-term degradation characteristics of the sensor in space.

Furthermore, our results can also be used to more accurately evaluate the sensitivity change of the SP. As seen in Fig. 11(b), the change of the sensor sensitivity of the SP based on the A16 observations showed a peak feature centered around $\lambda = 900$–1100 nm [19]. There are two possible reasons for the peak feature. One is that the peak feature around $\lambda = 900$–1100 nm is due to an error in the estimate of the A16 observations. Although the same target sites for the A16 observations were observed as much as possible, we cannot rule out the possibility that there are slight differences in the observation areas. In other words, some spatial variation in the A16 observation sites may result in the peak feature. Alternatively, we can also consider that the peak feature may reflect a significant change of the sensitivity of the SP in this wavelength range, because the boundary between VIS and NIR 1 detectors is located at 950–1000 nm, and a signal-to-noise ratio is relatively low at wavelengths around the boundary owing to the characteristics of the detectors [19]. In any case, it was difficult to determine the reason for this only from the data of the A16 observations. On the other hand, our results can provide a constraint on this issue. If this is due to the significant change of the sensitivity around 900–1100 nm, the difference in the S-ratio between P1 and P4 ($\delta\chi_4$) should also show a similar trend with the peak feature around $\lambda = 900$–1100 nm. However, our results exhibit a different trend, showing that the shorter the wavelength, the larger the decrease, especially for shorter VIS wavelengths. Thus, it is likely that the peak feature around $\lambda = 900$–1100 nm is due to spatial variation in the A16 observation sites. The results of the onboard lamp of the SP enable us to assess the accuracy of the results of the vicarious calibration using the A16 observations.

VI. Conclusion

We assessed the degradation of the radiance lamp in the internal calibration system of the SP onboard SELENE.

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We found that the S-ratio for VIS in the first period of $d <$ 250 days were ∼1.032–1.040, suggesting that the output of the radiance lamp increased at least ∼3.2–4.0% after launch. Then, the S-ratio after $d = 343$ days decreased with increasing d, and the total amount of the decrease from the first period ($d < 250$) to the last period ($d > 585$) was ∼1.1%–1.5%. For NIR 1, it was also shown that the S-ratio for $d < 250$ days were ∼1.035–1.046, suggesting that the output of the radiance lamp increased by more than ∼3%–4% after launch, which is consistent with the results for VIS. Then, the S-ratio decreased with increasing d after $d = 343$ days, and the total decrease from the first period to the last period was ∼1.1%–1.3%, which is also consistent with the results for VIS.

We also found that the temporal change based on the radiance lamp was larger than that for the sensor sensitivity of the SP derived from the A16 observations, indicating that the radiance lamp became degraded during the mission period. From the difference in the change between the radiance lamp and the A16 observations, we estimated that the degree of lamp degradation was up to 1% between the first period and the last period. Our results also showed that the degree of decrease in the S-ratio depended on λ, where the shorter the wavelength, the larger the decrease in the S-ratio, especially for shorter wavelengths in VIS. We demonstrated that the trend of the wavelength dependence can be accounted for by the decrease in temperature of the radiance lamp. In addition, the temperature decrease model with ΔT of about −5 K was shown to be consistent with the degradation of the radiance lamp data over about 1.5 years. In this case, we may estimate that the temperature decrease rate was about −3 K/yr.

Because a quantitative estimate of the degradation of the radiance lamp has not been well established for previous remote-sensing sensors, our results may provide important information for calibration methods using a radiance lamp for other optical sensors, such as ASTER, which uses lunar calibrations and has lamp monitoring data that spans over two decades [29], and HISUI which provides hyperspectral data and has lamp calibration systems [30].

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Satoru Yamamoto received the Ph.D. degree in science from Kobe University, Kobe, Japan, in 1999.

He is currently a Senior Researcher with the Research Institute of Geology and Geoinformation, Geological Survey of Japan, National Institute of Advanced Industrial Science and Technology, Tsukuba, Japan. He is involved in the Earth-observation remote-sensing projects of ASTER and HISUI and the Lunar Mission Project of SELENE/Kaguya.

Dr. Yamamoto is a member of the Japanese Society of Planetary Sciences, the Japan Geoscience Union, and the Remote Sensing Society of Japan.



Hiroki Mizuochi received the Ph.D. degree in environmental studies from the University of Tsukuba, Tsukuba, Japan, in 2017.

He is currently a Researcher with the Research Institute of Geology and Geoinformation, Geological Survey of Japan, National Institute of Advanced Industrial Science and Technology, Tsukuba. He is involved in the Earth-observation remote-sensing projects of ASTER and HISUI, and Pan-Arctic Water–Carbon Cycles JSPS Scientific Research.

Dr. Mizuochi is a member of the Remote Sensing Society of Japan, the Japan Society of Photogrammetry and Remote Sensing, and the Japan Geoscience Union.



Moe Matsuoka received the Ph.D. degree in science from Tohoku University, Sendai, Japan, in 2018.

She is currently a Researcher with LESIA, Observatoire de Paris, Paris, France. She is involved in the asteroid exploration missions of Hayabusa2 and MMX.

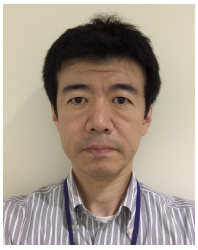
Dr. Matsuoka is a member of the Japanese Society of Planetary Sciences and the American Geophysical Union.



Koki Iwao received the Ph.D. degree in engineering from the University of Tokyo, Tokyo, Japan, in 2000.

He is currently a Group Leader of the Remote Sensing Research Group with the Research Institute of Geology and Geoinformation, Geological Survey of Japan, National Institute of Advanced Industrial Science and Technology, Tsukuba, Japan. He is involved in the Earth-observation remote-sensing projects of ASTER and HISUI.

Dr. Iwao is a member of the Remote Sensing Society of Japan, the Japan Society of Photogrammetry and Remote Sensing, and the Japan Association of Surveyors.



Satoshi Tsuchida received the B.S. degree in geology and the Ph.D. degree in mineral resources engineering from Waseda University, Tokyo, Japan, in 1986 and 1991 respectively.

He joined the Geological Survey of Japan (GSJ) in 1992 to engage in research in geological and environmental remote sensing and calibration of radiometers on satellites. He is currently a Principal Research Manager with the GSJ, National Institute of Advanced Industrial Science and Technology, Tsukuba, Japan.

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